

Graphical Network Simulator that can Run on a Raspberry Pi

Here is a cool tool for learning and testing in a simulated environment. Use this open source and free tool to design and test your own network with various network hardware devices and hosts, all on your Raspberry Pi. Read this article to know what it does

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GNS3 is a graphical network simulator which provides you a real lab environment for learning and testing complex networks. With this tool you do not

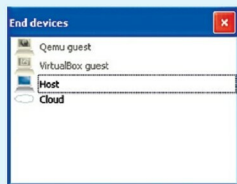


Fig 1: End devices

need to have dedicated network hardware devices. You can design and configure virtual networks on a traditional PC or a Raspberry Pi, and simulate in the closest possible way to how the real networks perform.

If you want to design and test your automation system network, GNS3 can just work fine for you. It allows you to run many virtual hosts with operating systems such as Linux, Windows, MacOS X or FreeBSD.

Integrated solution for network emulation

It uses the Dynamips, Pemu/Qemu and Dynagen emulators to run the very same operating systems as in

real networks, ensuring complete and accurate simulations. Dynamips is the core program that allows IOS emulation.

Other emulation programs, namely Qemu, Pemu and VirtualBox, are used to emulate Cisco ASA and PIX firewalls, Cisco IPS, Juniper routers as well as hosts (Linux, Windows, MacOS X and FreeBSD). GNS3 integrates all the emulation and allows you to work in a real lab environment to have your host talking to your routers.

Better graphical front-end for ease of use

The GNS3 software provides an intuitive graphical user interface (GUI) to design and configure virtual networks running on traditional PC hardware for multiple operating systems, including Windows, Linux and MacOS.

GNS3 serves as a graphical front-end for the network emulators. For instance, using network emulators such as Dynagen or Dynamips independently will require you to build your labs using configuration files, but using a graphical tool like GNS3 makes things a bit easier for you.

GNS3 runs on top of these emulators to create a more user-friendly and graphical environment. You can click and drag different routers to your screen, hit a start button and start learning and testing.

While GNS3 is developed in Py-

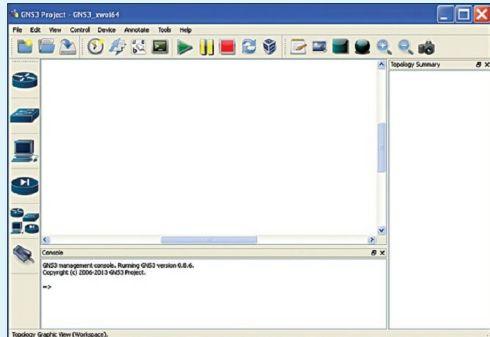


Fig 2: GNS3 main window